1. C
2. C
3. D
4. D
5. A
6. B
7. B
8. D
9. D
10. B
11. B
12. C
13. B
14. C
15. B
16. B
17. B

40

10

18. C

19. A

20. B

21. Supervised Learning is a type of machine learning that are used to train and predict data that are labeled.

22. Model M1 is not a good model, despite being able to get a 99% accuracy for train dataset it still fails at the testing dataset. This can occur to certain reasons:
- Lack of Dataset
- Bad feature Selection
- Too many outliers
- No Feature Engineering.

25. 2 major applications of data science:
- Heart - Failure Prediction
- Classifying of Tumor is Dormant or Malignant.

27. To find the outlier in the Data we could perform a standard deviation by using the mean as a middle point to determine which data is furthest, the ~~fad~~ furthest are the outliers.

28. Data pre-processing is a stage in the data science life-cycle. During this stage is where we prepare the dataset inorder to feed it to the model for ~~th~~ training and Testing. In this Stage, we perform:
- handling miss value
- Outlier Detection
- feature Selection
- feature Engineering

~~29.~~

29. Data normalization is required in K-NN, because we need to divide the dataset into class or group that has similar property. This will allow the k-NN model to precisely train with ~~table~~ labeled data and make better prediction.

31. k in k-means algorithms refers to the number of clusters that we will use to divide the dataset into when fiting the model.

32. A few hyperparameters of ML Algorithms:
- n-estimators
- mean
- media
- neighbors
- k - number of clusters
- Train-Test set ratio

34. Comment on the performance of the Classifiers
- Classifier 1: this is a good classifier, whereas there are no mis-classification ~~and it remains a~~, able to sep separate the data properly and has a decent gap between the linear boundaries.

34. - Classifier 2: this ~~is~~ ~~a~~ not a good classifier. There seems to be 2 different types of data within the linear boundarres gap. ~~But it~~ Despite having a deeent gap lenght, it stills mis-classify data properly.

- Classifier 3: this is an ok classifier. Even though there are no ~~mis~~ mis-classification with the data, this classifier has a rather small gap between the linear boundarres.

23. Regression Analysis is the monitoring of the relationship between variables with contrinous value.

24. 2 challenges of Machine Learnng:
- Data: This is a major problem, even with the best ML Algorithms with bad data the output would also be bad.

- Uncertainties: Despite having enough data to train the model, this still could not determine the performance. Since uncertainties may occur, it may effect it's performance.

	Structured Data	Semi-Structured Data
26.	- Are in tabular format	- Not in tabular format
	- Has all features that builds up conclusion	- Has features all mixed
	- format of Excel, CSV	- Format of image, video, sound

30\. Continuous Attributes are discretized by calculating the mean of the Attribute and measure the distance from one another.

33\. Gradient Descent is the derivative value of the dependent variable, where it alters the final output of every calculation with SVM.

35\. Radial Basis Function Kernel work in SVM by measuring and comparing the Distance of the data from the linear boundaries.